

Practical 2a. Testing IID and Gaussian Characters

1. Generate samples of length n of different IID noises using the following instructions.

- `rnorm(n, mean=, sd=)`
- `runif(n, min=, max=)`
- `rexp(n, rate=)`
- `rchisq(n, df=)`
- `rpois(n, lambda=)`
- `rt(n, df=)`

Observe what happens when you move the parameters.
For different distributions, do for example:

- `n<-1000`
- `x<-rexp(n,rate=2)`
- `plot(x,type="l")`
- `hist(x,k)`

Here k denotes the number of classes for the histogram. Test with other distributions writing “distributions” in the help utility

2. Verify the IID character of any sample of the previous exercise. Use the Ljung-Box test with the instruction.

- `Box.test(x, lag=h, type="Ljung")`

For example:

- `n<-10000`
- `k<-3`
- `x<-rt(n, df=k)`
- `plot(x, type="l")`
- `Box.test(x, lag=h, type="Ljung")`

What happens if we do the same for a random walk?

Try with different lags, from $h=1$ to $h=\log(n)$.

Practice 2b: Exponential smoothing and Holt-Winters forecasting

1. Generate an IID noise of length n and find its exponential smoothing series.

The following piece of code generates the Holt Winters smoothing for stationary series with no trend and nor seasonality.

```
➤ n<-100
➤ x<-rnorm(n)
➤ plot(x)
➤ a<-0.1
➤ y<-HoltWinters(x,alpha=a,beta=F,gamma=F)
➤ plot(y)
```

Change n. Change parameter “a” between 0 and 1. Simulate other IID noises with other laws. If do not prefix “a” the parameter is estimated.

2. Apply Holt Winters smoothing to real data with trend and seasonality.

```
➤ x<-AirPassengers
➤ a<-0.2
➤ b<-0.3
➤ c<-0.8
➤ y<-HoltWinters(x,alpha=a,beta=b, gamma=c)
➤ plot(y)
```

Change the series and use for example $\log(x)$, $\text{diff}(\log(x))$, $\text{diff}(x,12)$, etc.

Change the values of parameters a, b and c

3. Holt Winters prediction:

Type in the following commands:

```
➤ x<-AirPassengers
➤ a<-0.3
➤ b<-0.1
➤ c<-0.8
➤ y<-HoltWinters(x,alpha=a,beta=b,gamma=c)
➤ plot(y)
➤ z<-predict(y,n.ahead=4*12)
➤ plot(y,z,lty=1:2)
```

3. If a, b, c are not prefixed we can do

```
➤ x<-AirPassengers
➤ y<-HoltWinters(x)
➤ y$alpha
➤ y$beta
➤ y$gamma
➤ plot(y)
➤ z<-predict(y,n.ahead=4*12)
➤ plot(y,z,lty=1:2)
```

Observe the estimations are $a=0.248$, $b=0.034$ and $c=1$.

5. Another example: wine sales in Australia.

- `x<-wineind`
- `y<-HoltWinters(x)`
- `plot(x)`
- `y$alpha`
- `y$beta`
- `y$gamma`
- `plot(y)`
- `z<-predict(y,n.ahead=4*12)`
- `plot(y,z,lty=1:2)`

Do the same for other examples.